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Integrating edge features and complementary attention mechanism for drug response prediction

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ABSTRACT

Predicting drug response in cancer cell lines is a vital field in precision medicine, supporting personalized treatment planning, optimizing drug selection, and enhancing the accuracy and effectiveness of cancer therapies. Although graph neural network-based models for drug response prediction have made significant progress in performance, they often focus solely on learning node embeddings while overlooking adjacency relationships between cell lines, limiting the model's ability to capture inter-cell line adjacency information. To address this limitation, we propose a novel model that constructs edge features by measuring similarity between cell lines and their k-nearest neighbors, integrating these edge features with a node-edge complementary attention mechanism. This approach enables the model to dynamically incorporate node and edge information, achieving complementary and collaborative feature learning. Such a design substantially improves the accuracy and biological interpretability of drug response prediction. Furthermore, to enhance the independence and complementarity of node and edge features, we introduce a complementary loss mechanism in the model and design a topology updating module that performs dynamic feature updates via neighborhood aggregation, effectively capturing and utilizing multi-omics data. We conduct comprehensive experiments on the Genomics of Drug Sensitivity in Cancer and the Cancer Cell Line Encyclopedia, which contains various diseases such as esophageal carcinoma, stomach adenocarcinoma, colon adenocarcinoma and rectal adenocarcinoma, the results demonstrate that our model outperforms current state-of-the-art methods in cancer drug response prediction.

1. Introduction

Cancer remains one of the leading causes of death worldwide, presenting significant treatment challenges due to its complex molecular characteristics and high heterogeneity [1–4]. The same type of cancer can vary dramatically in gene expression, mutation profiles, and molecular characteristics across different patients, resulting in vastly different responses to the same drug. This high level of individuality underscores the importance of personalized drug response prediction in cancer treatment [5–7]. By analyzing multi-omics data from cell lines or patients, it is possible to identify unique genetic and molecular features that inform individualized treatment plans, optimize drug combinations, enhance therapeutic efficacy, and reduce side effects.

Conducting experimental tests of multiple treatment strategies on cancer patients is not only time-consuming and costly but also poses potential risks, making it infeasible to test broadly on patients directly. Instead, drug response datasets allow for in vitro modeling and validation of cell line and drug responses, effectively simulating patient reactions to drugs and providing a scientific basis for personalized treatment. Commonly used datasets include the GDSC and the CCLE, which encompass genomic data, multi-omics features, and drug sensitivity measurements across various cancer cell lines. These datasets enable models to effectively learn the patterns of cancer cell responses to different drugs, aiding in the development of precision cancer therapies.

Furthermore, drug response is influenced not only by genetic mutations but also by complex biological interactions between cell lines. For example, gene expression similarity between cancer cell lines often correlates with shared drug sensitivity. Previous studies have shown that cell lines with similar oncogenic pathways, such as the activation of the PI3K/AKT/mTOR or MAPK signaling pathways, tend to exhibit comparable responses to targeted therapies [8,9].

In the field of drug response prediction, with the rapid development of bioinformatics and machine learning, many methods have been

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proposed and widely applied to the drug response prediction task. These methods can typically be categorized into the following types: regression models, classification models, and link prediction models. Regression models are commonly used for continuous output problems in drug response prediction, with the main goal of predicting the sensitivity or effect size of a cell line to a drug. Early regression models utilized methods like LASSO [10], elastic net [11] and ridge regression [12], to estimate continuous variables like IC50 concentrations. Regression models are simple and easy to implement but lack the capability to model non-linear relationships in data. Classification models, on the other hand, are primarily used to categorize drug responses into classes such as sensitive, resistant, or neutral. Traditional methods include extracting features from samples and using algorithms like support vector machines [13] and random forests [14,15] for classification. In contrast, deep learning models, such as deep neural networks [16], convolutional neural networks [17], and autoencoders [18], can extract high-level features from the raw data of cell lines and drugs, and integrate these features through multiple layers of non-linear transformations for prediction [19,20].

Link prediction models are primarily used to infer potential connections in a graph, specifically predicting the relationship between drugs and cell lines (e.g., whether a drug is effective). Link prediction methods typically use existing information in a graph to infer missing edges. With the rise of Graph Neural Networks (GNNs) [21,22] in bioinformatics, link prediction models often employ graph structures to model the complex relationships between drugs and cell lines [23]. GNNs treat drugs and cell lines as nodes in a graph, and the similarity or interaction between them is represented as edges in the graph. Zhang et al. [24] developed a heterogeneous network that includes cell systems, drugs, and their interactions. They used an information flow-based algorithm to predict potential drug responses in these cell systems. Although this method enhances prediction capabilities by utilizing various heterogeneous data sources, it requires extensive prior knowledge. Liu et al. [25] developed an innovative drug response prediction technique that combines a contrastive learning framework with Graph Convolutional Networks (GCNs). This approach improves the model's ability to distinguish between negative and positive samples, thereby enhancing the generalization performance. Peng et al. [26] integrated multi-omics data to construct heterogeneous networks and used GCNs to obtain embedding features for both cell lines and drugs. In NIHGCN [27], they combined Graph Convolutional Networks and neighborhood information layers to classify drug responses in cell lines. Xu et al. [28] proposed a dual-branch architecture with dual relationships that facilitates the independent learning and interactive fusion of cell line and drug attribute information as well as their associated relationships.

Traditional machine learning and deep learning models have made significant progress in this field, but they still face several limitations: (1) Limited consideration of cell line similarity: Many existing methods primarily focus on learning individual node embeddings without capturing relationships between similar cell lines. Although, biological studies indicate that cell lines with similar genetic backgrounds and pathway activities often exhibit similar drug responses, making it crucial to incorporate this relational information. (2) Lack of feature complementarity: Traditional GNN-based methods rely heavily on node features while overlooking edge features that encode interactions between cell lines, which limits their ability to effectively model the complementary information between nodes and edges, ultimately reducing their capacity to fully utilize multi-omics data for drug response prediction. (3) Heterogeneity in multi-omics integration: Different omics data types exhibit distinct statistical properties, biological meanings, and sparsity patterns. For instance, the characteristically sparse nature of mutation data (typically exhibiting <1% mutation rates) presents significant integration challenges when combined with dense expression profiles. Existing methods often treat all omics layers uniformly, failing to account for their intrinsic disparities in scale, biological context, and data density.

How Our Model Addresses These Issues: To overcome these challenges, we propose ECACDR, which introduces: (1) Edge features based on cell line similarity to capture biologically meaningful relationships. (2) A complementary attention mechanism to enable dynamic feature fusion of node and edge information. (3) A topology-aware update module to refine cell line representations via neighborhood aggregation, improving generalization across datasets.

Overall, our main contributions are as follows:

- We propose a complementary attention mechanism, which allows the model to consider the introduced edge features while learning node features, promoting complementary and collaborative information. This effectively addresses the previous models' neglect of the relationships between similar cell lines, thereby improving the accuracy and interpretability of drug response prediction.
- We construct a topology-aware updater that combines the current node features with neighborhood aggregation, enhancing local context capture, supporting multi-omics integration, and maintaining the feature uniqueness required for accurate predictions.
- We conduct extensive comparative experiments on nine state-of-the-art algorithms. The experimental results demonstrate that our algorithm, ECACDR, achieves superior performance on the GDSC and CCLE datasets.

2. Related work

Predicting cancer drug response is a critical task in precision medicine, and recent advances have significantly improved performance in this field.

2.1. Deep learning drug response prediction

Traditional machine learning models, such as Random Forest and Support Vector Machines, struggle with high-dimensional omics data and require extensive feature engineering. Deep learning approaches have demonstrated superior performance by automatically extracting meaningful representations. Hossein et al. proposed a deep neural network-based late integration approach for multi-omics data fusion [18], while Hira et al. [29] introduced a variational autoencoder to address the high dimensionality of multi-omics data. Jiang et al. designed an end-to-end model that integrates transformers for drug representation learning with multilayer neural networks to predict anticancer drug responses from transcriptomics data [30]. DeepDR [31] provides a library with multiple encoding techniques, enabling flexible model construction. Although these approaches achieve promising results, they primarily focus on independent omics features without considering relational structures between cell lines.

2.2. Graph-based methods

Graph Neural Networks (GNNs) have emerged as powerful tools for modeling drug-cell line interactions due to their ability to capture topological relationships in biological data. Ahmed et al. successfully applied GNNs for gene representation learning [32], and Peng et al. [26] constructed a heterogeneous cell line-drug network, employing graph convolutional operations to update features based on network topology. Liu et al. [25] further enhanced GCN-based drug response prediction by introducing a contrastive learning framework, improving generalization and accuracy.

Despite the rapid progress in GNN-based models, most existing methods focus primarily on node features, such as drug molecular structures and omics profiles, while overlooking edge features that represent relationships between cell lines. As a result, these models struggle to effectively capture biologically meaningful interactions.

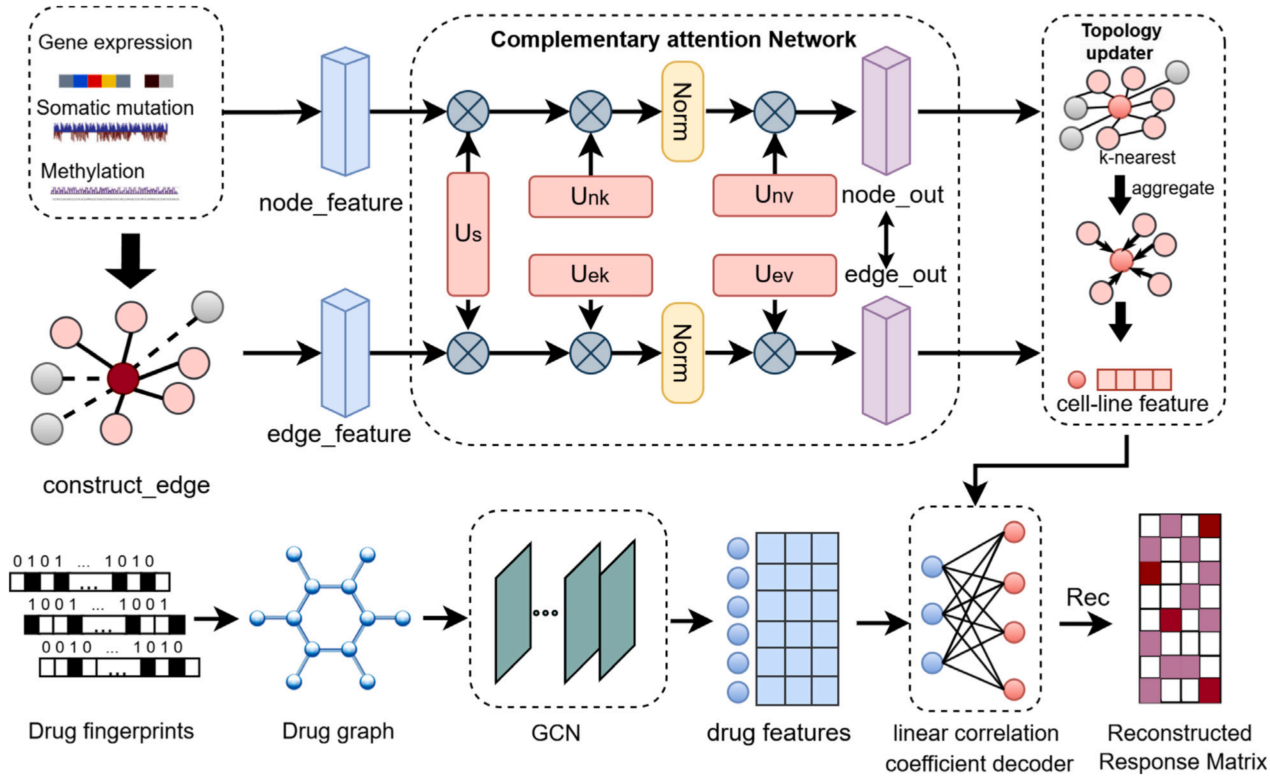


Fig. 1. The model constructs cell line embeddings by integrating node and edge features through a complementary attention module, with losses applied to both node_out and edge_out. A topology-aware updater further refines node embeddings through neighborhood aggregation. A Graph Convolutional Network (GCN) is applied to extract a latent representation of each drug. Finally, the interaction between drugs and cell lines is modeled using a linear correlation coefficient decoder to predict drug response.

2.3. Knowledge-aware models

Recent studies have explored knowledge graphs (KGs) to enrich molecular interaction prediction. KG-MTL [33] integrates knowledge graphs and molecular graphs through multi-task learning, achieving improved drug-target interaction predictions. Its primary focus is on drug-protein interactions and does not incorporate dynamic graph structures or cell line similarities, both of which are essential for drug response prediction. To improve robustness, BioKDN [34] applies knowledge graph denoising techniques to eliminate noisy interactions in biomedical knowledge graphs. This approach enhances the reliability of molecular interaction predictions but does not specifically target drug response prediction at the cell line level, limiting its applicability in precision medicine.

2.4. Remaining challenges

Deep learning, graph-based models, and knowledge graphs have collectively advanced the field of drug response prediction. Despite these improvements, significant challenges remain in the ability to model dynamic cell line relationships, integrate complementary information from node and edge features, and enhance the biological interpretability of predictions.

Existing methods predominantly rely on node features, such as omics data and drug descriptors, while paying little attention to the relationships between cell lines. The absence of explicit edge feature modeling results in information loss, particularly in capturing biologically meaningful similarity patterns. Many approaches also struggle to balance the independence and integration of node and edge features, leading to suboptimal interpretability of model predictions.

To address these limitations, the proposed model introduces dynamically constructed edge features that capture cell line relationships based on biological similarity. A complementary attention mechanism ensures that node and edge features remain independent yet contribute collaboratively to the learning process. Additionally, a topology-aware update module refines node representations through neighborhood aggregation, reflecting dynamic biological contexts and improving the integration of multi-omics data. These innovations enhance both predictive accuracy and biological interpretability, offering a more comprehensive framework for cancer drug response prediction.

3. Proposed method

In this section, we begin by presenting the framework of the ECACDR. Subsequently, a detailed description of each module within the model is provided.

3.1. Overview of ECACDR

Fig. 1 illustrates the workflow of the ECACDR model. First, based on data such as gene expression, mutations, and methylation, the similarity between cell lines is calculated, and edge features are constructed for each cell line based on its k-nearest neighbors. Then, node features and edge features are input into the complementary attention module, where complementary learning produces node embeddings and edge embeddings. Next, the topology feature update module is applied for multi-omics data fusion [35–38] to obtain the final cell line embedding. Finally, the cell line embeddings are combined with the drug embeddings extracted through graph convolution operations, and then utilize a linear correlation coefficient decoder for drug response prediction.

3.2. Detailed module of ECACDR

3.2.1. Construction of edge features

In drug response prediction, understanding the similarities and relationships between cell lines is crucial for capturing the biological basis of drug sensitivity. For instance, similar gene expression or mutation patterns may result in comparable responses to the same drug. We introduce edge features between similar cell lines to smooth out subtle differences between neighboring nodes, thereby mitigating the impact of noise or extreme feature values on the learning process. Initially, we separately process the gene expression, mutation, and methylation data of cell lines to generate unified D-dimensional embeddings for each type of omics feature [25,39]:

$$h_i^s = \mathrm{Re}LU\left(f_s\left(p_i^s\right)\mathbf{W}_b + b_s\right), \tag{1}$$

where s represents different types of omics data, p_i^s denotes the raw data for each omics type, f_s refers to a fully connected network, and $\mathbf{W}_s$ and b_s represent the weight matrix and bias vector, respectively.

After obtaining the unified D-dimensional features for the various omics data of cell lines, for any given pair of cell lines (c_i, c_j), the kernel matrix for each omics type can be computed using a cosine kernel function as follows:

$$\mathbf{R}_{ij}^s = k\left(h_i^s, h_j^s\right) = \frac{h_i^s \cdot \left(h_j^s\right)^\top}{\|h_i^s\|\,\|h_j^s\|}. \tag{2}$$

Subsequently, the edge set is obtained by selecting the k-most similar neighbors for each cell line using the k-NN method:

$$\varepsilon = \{(i,j)\,|\,Neighbors\,(i)\}, \tag{3}$$

where $Neighbors\,\{i\}$ represents the k-most similar neighbors of cell line i. The feature of each edge is generated as the arithmetic mean of the features of the connected nodes:

$$e_{ij} = \frac{h_i + h_j}{2}, \forall (i,j) \in \varepsilon. \tag{4}$$

Through e_{ij}, we can form an edge feature matrix $\mathbf{E}$, where each row represents a feature balance point constructed between two adjacent samples. This central feature effectively retains the key information from both samples while mitigating the influence of extreme values from individual features on subsequent computations or model learning. The biological significance of edge feature construction lies in its ability to capture inherent similarities between cell lines, which may be indicative of shared drug responses. For instance, breast cancer cell lines overexpressing HER2, such as BT-474 and SK-BR-3, are known to be highly responsive to Trastuzumab [40,41]. By modeling similarity through edge features, our approach enables the identification of such clinically relevant patterns.

3.2.2. Complementary attention mechanism module

From a biological perspective, drug response is not solely influenced by the individual characteristics of cell lines, interactions between cell lines also play a significant role in drug response. Traditional self-attention mechanisms (e.g., those used in Transformer models) compute attention weights based solely on node features, typically by measuring similarities between elements. While this approach performs well in many tasks, it lacks the capacity to model edge information between neighboring nodes in biologically relevant tasks such as drug response prediction, thereby overlooking biological patterns.

Inspired by Liang et al. [42], we introduce a Complementary Attention Mechanism to further enhance the synergy between node features and edge features, rather than solely relying on self-attention mechanisms to process inter-node relationships. This approach improves the model's understanding of cell lines, as well as their similarities and interactions.

Starting with the input node features $\mathbf{X}$, the self-attention mechanism in Transformers [43] projects $\mathbf{X}$ into three matrices: the query matrix $\mathbf{Q}$, key matrix $\mathbf{K}$, and value matrix $\mathbf{V}$. The self-attention formula can be expressed as follows:

$$\begin{aligned} \mathbf{A}_{\mathrm{self}} &= \operatorname{soft\,max}\left(\frac{\mathbf{QK}^\top}{\sqrt{d_{out}}}\right), \\ \text{Self-Attn}\,(\mathbf{X}) &= \mathbf{A}_{\mathrm{self}}\mathbf{V}, \end{aligned} \tag{5}$$

where d_{out} represents the length of a row vector in $\mathbf{Q}$. In contrast, we introduce an external unit to compute attention between the input features and the external unit:

$$\begin{aligned} \mathbf{A}_{\mathrm{com}} &= \mathrm{norm}\left(\mathbf{XU}^\top\right) \in \mathbb{R}^{\mathrm{n}\times\mathrm{S}}, \\ \text{Com-Attn}\,(\mathbf{X}) &= \mathbf{A}_{\mathrm{com}}\mathbf{U} \in \mathbb{R}^{\mathrm{n}\times\mathrm{d}}. \end{aligned} \tag{6}$$

Here, $\mathbf{U} \in \mathbb{R}^{\mathrm{S}\times\mathrm{d}}$ is a learnable parameter that is independent of the input data. It can be regarded as an external unit with S nodes, serving as shared memory for both input node features and edge features. The external unit undergoes two layers of linear transformations to adjust the representation space of the features. In self-attention, A_{self} represents the similarity between nodes, whereas in complementary attention, A_{com} represents the similarity between input graph nodes and the external unit. To mitigate instability caused by feature scale variations and to emphasize directional information, a dual normalization technique [44] is applied in the complementary attention module. The dual normalization, involving both row and column normalization, can be expressed as:

$$\begin{aligned} \tilde{\mu}_{i,j} &= \left(\mathbf{XU}^\top\right)_{\mathrm{i,j}}, \\ \hat{\mu}_{i,j} &= \frac{\exp\left(\tilde{\mu}_{i,j}\right)}{\sum_{\mathrm{k=0}}^{\mathrm{n}} \exp\left(\tilde{\mu}_{k,j}\right)}, \\ \mu_{i,j} &= \frac{\hat{\mu}_{i,j}}{\sum_{\mathrm{k=0}}^{\mathrm{S}} \hat{\mu}_{i,k}}. \end{aligned} \tag{7}$$

We assign distinct external units for node features and edge features. The node embeddings and edge embeddings after processing with the complementary attention mechanism can be expressed as follows:

$$\begin{aligned} \mathbf{H}_{\mathrm{out}} &= \mathrm{norm}\left(\mathbf{HU}_{\mathrm{S}}\mathbf{U}_{\mathrm{nk}}^\top\right)\mathbf{U}_{\mathrm{nv}}, \\ \mathbf{E}_{\mathrm{out}} &= \mathrm{norm}\left(\mathbf{EU}_{\mathrm{S}}\mathbf{U}_{\mathrm{ek}}^\top\right)\mathbf{U}_{\mathrm{nev}}. \end{aligned} \tag{8}$$

Here, $\mathbf{U}_S \in \mathbb{R}^{\mathrm{d}\times\mathrm{d}}$ is a shared unit connecting node features and edge features. $\mathbf{U}_{nk}, \mathbf{U}_{nv} \in \mathbb{R}^{\mathrm{S}\times\mathrm{d}}$ are exclusive external key–value units for node features, while $\mathbf{U}_{ek}, \mathbf{U}_{ev} \in \mathbb{R}^{\mathrm{S}\times\mathrm{d}}$ are external key–value units for edge features, $\mathbf{H}$ and $\mathbf{E}$ represent the node feature matrix and edge feature matrix, respectively.

3.2.3. Multi-omics topology update and fusion module

To achieve the integration of multi-omics information, better capture the local graph topology among nodes, and enhance the expressiveness of features, we designed a topology update module based on the distances between nodes. The topology-aware update module addresses multi-omics heterogeneity through: (1) Omics-specific neighborhood aggregation: For each omics type, cell line features are first updated independently by aggregating information from k-nearest neighbors within the same omics layer, preserving type-specific patterns. (2) Concatenation-based fusion: The updated omics-specific features are concatenated to form the final cell line representation.

First, the outputs of each omics dataset obtained through the complementary attention module are concatenated:

$$\mathrm{q} = \left[\mathrm{h}_{\mathrm{out}}^{\mathrm{u}} \| \mathrm{h}_{\mathrm{out}}^{\mathrm{e}} \| \mathrm{h}_{\mathrm{out}}^{\mathrm{m}}\right], \tag{9}$$

where $\|$ represents the vector concatenation operator, while $\mathrm{h}_{\mathrm{out}}^{\mathrm{u}}$, $\mathrm{h}_{\mathrm{out}}^{\mathrm{e}}$, and $\mathrm{h}_{\mathrm{out}}^{\mathrm{m}}$ denote the embedding outputs of mutation, gene expression, and methylation data through the complementary attention module, respectively. For all nodes, the relationships between node pairs are computed using the Euclidean distance:

$$\mathrm{distance\,(i,j)} = \sqrt{\sum_{\mathrm{t}}^{3\times\mathrm{D}} \left(\mathrm{q}_{\mathrm{i,t}} - \mathrm{q}_{\mathrm{j,t}}\right)^2}, \tag{10}$$

where $\mathrm{q}_{\mathrm{i,t}}$ and $\mathrm{q}_{\mathrm{j,t}}$ represent the values of nodes i and j in feature dimension t.

Based on the calculated distances, the K-nearest neighbor set for each node is selected:

$$\mathcal{N}_i = \{j\,|\,j \in \mathrm{top}_K(\mathrm{distance}(i,j))\}. \tag{11}$$

For each node i, the topological feature is constructed by calculating the mean of the features of its K-nearest neighbors:

$$\hat{q}_i^{\text{topo}} = \frac{1}{K}\sum_{j\in\mathcal{N}_i} q_j. \tag{12}$$

The edge feature e_{ij} between a node i and its neighboring node j is computed using the following formula:

$$e_{ij} = q_j + \hat{q}_i^{\text{topo}}. \tag{13}$$

Finally, the aggregation of node features is enhanced as follows:

$$z_i = q_i + \frac{1}{K}\sum_{j\in\mathcal{N}_i} e_{ij}. \tag{14}$$

From the final embedding z of each cell line, we can obtain the final cell line embedding matrix $\mathbf{H_c}$.

3.2.4. Multi-task linear correlation coefficient decoder

For the cell lines, we obtain their feature matrix based on the aforementioned complementary attention and topology update modules. For the drugs, we can represent their molecular structure as a graph, where the nodes represent atoms, and the edges represent chemical bonds. The drug graph can be represented as $\{\mathcal{G}_j = (\mathbf{X}_j^d, \mathbf{A}_j^d)|_{j=1}^{N_D}\}$, where $\mathbf{X}_j^d$ and $\mathbf{A}_j^d$ denote the feature matrix and adjacency matrix of the jth drug, respectively. Similar to previous methods [25], we use multi-layer GCNs to capture the latent representations between drugs and obtain their embedding matrix $\mathbf{H_d}$.

Similar to [45], we then reconstruct the cell line-drug association matrix using a multi-task linear correlation coefficient decoder, performing regression prediction and similarity reconstruction. The linear correlation coefficient is given by the following formula:

$$Corr(x_i, x_j) = \frac{(x_i-\mu_i)(x_j-\mu_j)^T}{\sqrt{(x_i-\mu_i)(x_i-\mu_i)^T}\sqrt{(x_j-\mu_j)(x_j-\mu_j)^T}} \quad , \tag{15}$$

where x_i and x_j represent the feature vectors of the cell line i and the drug j, and μ_i and μ_j are the averages of them respectively. Then, reconstruct the association matrix as:

$$\hat{\mathbf{A}} = \varphi\left(\text{Corr}\left(\mathbf{H_c}, \mathbf{H_d}\right)\right), \tag{16}$$

where φ is the sigmoid activation function. Binary cross-entropy is utilized as the loss function for prediction:

$$\ell_{bin}\left(\mathbf{A},\hat{\mathbf{A}}\right) = -\frac{1}{m\times n}\mathbf{M}_{ij}\left[\mathbf{A}_{ij}\ln\left(\hat{\mathbf{A}}_{ij}\right) + \left(1-\mathbf{A}_{ij}\right)\ln\left(1-\hat{\mathbf{A}}_{ij}\right)\right]. \tag{17}$$

where $\mathbf{A}$ represents the original cell line-drug association matrix. We estimate the drug-cell line response concentration (IC50 values) by computing the linear correlation coefficient between the final embeddings. Ultimately, the IC50 response matrix can be reconstructed:

$$\hat{\mathbf{A}}_r = Corr\left(\mathbf{H_c}, \mathbf{H_d}\right). \tag{18}$$

We utilize mean square error as the loss function for this regression task:

$$\ell_{\text{reg}}\left(\mathbf{A}_r, \hat{\mathbf{A}}_r\right) = -\frac{1}{m\times n}\sum_{\text{i,j}}\mathbf{T}_{ij}\left(\mathbf{A}_{r(ij)} - \hat{\mathbf{A}}_{r(ij)}\right)^2, \tag{19}$$

where T serves as an indicator matrix that aids in identifying specific cell line-drug associations.

The similarity of cell line or drug can be estimated by computing the linear correlation coefficient between the final embeddings of cell lines $\mathbf{H_c}$ or drug embeddings $\mathbf{H_d}$. This process yields reconstructed the similarity matrices of them. Subsequently, we calculate the reconstruction loss for cell line similarity ℓ_c and drug similarity ℓ_d using mean square error.

$$\ell_{\text{c}}\left(\mathbf{S}_c, \hat{\mathbf{S}}_c\right) = -\frac{1}{\text{m}\times\text{n}}\sum_{\text{ij}}\mathbf{T}_{ij}\left(\mathbf{S}_{c(ij)} - \hat{\mathbf{S}}_{c(ij)}\right)^2, \tag{20}$$

$$\ell_{\text{d}}\left(\mathbf{S}_d, \hat{\mathbf{S}}_d\right) = -\frac{1}{\text{m}\times\text{n}}\sum_{\text{i,j}}\mathbf{T}_{ij}\left(\mathbf{S}_d - \hat{\mathbf{S}}_d\right)^2, \tag{21}$$

$$\ell_{sim} = \ell_c\left(\mathbf{S}_c, \hat{\mathbf{S}}_c\right) + \ell_d\left(\mathbf{S}_d, \hat{\mathbf{S}}_d\right), \tag{22}$$

where $\hat{\mathbf{S}}_c$ and $\hat{\mathbf{S}}_d$ respectively represent the reconstructed cell line and drug similarity matrices obtained through correlation coefficient calculation, $\mathbf{S}_c$ and $\mathbf{S}_d$ are the cell line and drug similarity matrices, respectively, computed based on gene expression profiles and drug molecular fingerprints.

The complementary loss is achieved by calculating the cosine similarity between the node features and edge features after the complementary attention mechanism. Combining the complementary learning loss and the multi-task joint loss, the final loss can be obtained:

$$L_{total} = \theta_b\ell_{bin} + \theta_r\ell_{reg} + \theta_s\ell_{sim} + \theta_c\ell_{com}, \tag{23}$$

where $\theta = \{\theta_\text{b}, \theta_\text{r}, \theta_\text{s}, \theta_\text{c}\}$ is used to control the weights, balancing their importance. The pseudocodes of ECACDR are illustrated in Algorithm 1.

3.3. Computational complexity analysis

We analyze the key components of our model in terms of computational efficiency:

Edge Feature Construction (k-NN similarity computation) : Complexity: $\mathcal{O}(n^2d)$, where n is the number of cell lines and d is the feature dimension. This step is performed only once during preprocessing and does not impact runtime inference. Complementary Attention Mechanism: Complexity: $\mathcal{O}(n^2d)$ (similar to self-attention mechanisms in transformers). Since attention is applied to a sparsely connected graph (instead of a fully connected one), computational overhead is controlled. Topology-Aware Updater: Complexity: $\mathcal{O}(nkd)$, where k is the number of nearest neighbors considered. This is significantly more efficient than fully connected GNNs, which require $\mathcal{O}(n^2d)$ computations.

Although ECACDR introduces additional computational steps, its structured design ensures that: Most expensive operations (edge feature construction) occur during preprocessing, reducing inference time. Sparse graph operations and localized updates keep computational costs scalable, making the model practical for large-scale drug response prediction tasks.

4. Experiments

4.1. Implementation details

We implemented the model code using the PyTorch framework and conducted experiments on an NVIDIA GeForce RTX 3090 GPU, employing the Adam optimizer to minimize the loss function. In the experiments, we partitioned the drug response data from the GDSC and CCLE datasets into training, validation, and test sets with a ratio of 60%, 20%, and 20%, respectively. Ten random seeds were selected for repeated experiments, and the parameters yielding the highest AUC in cross-validation were chosen. During the extraction and processing of cell line and drug features, their dimensions were uniformly set to 100. In the complementary attention module, two attention heads were used, with the unit dimension set to 50. The weight parameters for the losses are as follows: $\theta_b = 0.75$, $\theta_r = 0.25$, $\theta_s = 0.25$, and $\theta_c = 1$, The learning rate is set to 0.0005.

Algorithm 1: ECACDR

Input: Gene expression matrix $\mathbf{P}^{\mathrm{e}}$, somatic mutation matrix $\mathbf{P}^{\mathrm{u}}$, methylation matrix $\mathbf{P}^{\mathrm{m}}$, drug fingerprint data, cell line-drug association matrix $\mathbf{A}$, parameters θ_b, θ_r, θ_s, θ_c, lr, $epoch$, k.

Output: Reconstructed cell line-drug association matrix $\hat{\mathbf{A}}$, reconstructed IC50 response matrix $\hat{\mathbf{A}}_r$, reconstructed cell line and drug similarity matrices $\hat{\mathbf{S}}_c$ and $\hat{\mathbf{S}}_d$.

/***Preprocessing***/;
Process omics data into D-dim features via Eq. (1);
Generate drug molecular graphs (atom features + adjacency matrix);
for $i = 0$ ***to*** $epoch - 1$ **do**
  /***Edge Feature Construction:***/;
  for *each cell line pair* (i, j) **do**
    Compute $\mathbf{R}^s_{ij}$ via $\mathbf{R}^s_{ij} = \frac{h^s_i \cdot (h^s_j)^\top}{\|h^s_i\|\|h^s_j\|}$ (Eq. (2))
  end
  Select K-nearest neighbors via $\varepsilon = \{(i, j)|\text{Neighbors}(i)\}$ (Eq. (3));
  Generate edge features $e_{ij} = \frac{h_i + h_j}{2}$ (Eq. (4));
  /***Complementary Attention:***/;
  $\mathbf{H}_{\mathbf{out}}, \mathbf{E}_{\mathbf{out}} = \mathrm{ComAttn}(\mathbf{H}, \mathbf{E})$ (eqs. (6) to (8));
  /***Topology Update:***/;
  $\mathbf{H}_{\mathbf{c}} = \mathrm{TopoUpt}\left(\mathbf{H}_{\mathbf{out}}, \mathbf{E}_{\mathbf{out}}\right)$ (eqs. (9) to (14));
  /***Drug Embedding via GCN:***/;
  for *each drug graph* $\mathcal{G}_j$ **do**
    $\mathbf{H}_d = \mathrm{MultiLayerGCN}(\mathbf{X}^d_j, \mathbf{A}^d_j)$
  end
  /***Multi-task Decoder:***/;
  Compute $\mathrm{Corr}(\mathbf{H}_c, \mathbf{H}_d)$ via Eq. (15);
  $\hat{\mathbf{A}} = \sigma(\mathrm{Corr})$ (Eq. (16));
  $\hat{\mathbf{A}}_r = \mathrm{Corr}(\mathbf{H}_c, \mathbf{H}_d)$ (Eq. (18));
  Reconstruct similarity matrices $\hat{\mathbf{S}}_c, \hat{\mathbf{S}}_d$ via Eqs. (20) and (21);
  $L_{\text{total}} = \theta_b L_{\text{bin}} + \theta_r L_{\text{reg}} + \theta_s L_{\text{sim}} + \theta_c L_{\text{com}}$ (Eq. (23));
  Updating parameters by gradient descent and backpropagation.
end
Return $\hat{\mathbf{A}}$, $\hat{\mathbf{A}}_r$, $\hat{\mathbf{S}}_c$, $\hat{\mathbf{S}}_d$.;

4.2. Materials

We used two datasets, GDSC and CCLE, to evaluate our model. For the cell lines, we collected their genomic mutation, gene expression, and methylation data. In the gene mutation data, the values 0 and 1 represent whether a mutation occurred at a specific position [46]. The gene expression data is represented as a 697-dimensional feature vector, with the values being the log-normalized TPM values collected. The methylation data is represented as an 808-dimensional feature vector. Finally, the features from each omics type were standardized to 100 dimensions. For the drug-related information, the PubChem database [47] provides the substructure fingerprint profiles of the drugs. The GDSC dataset contains two tables: one, named TableS4 A, records the IC50 values for cell line-drug combinations, while the other, TableS5C, records the drug sensitivity thresholds. These two tables help infer drug sensitivity and resistance. The CCLE dataset provides information on drug targets, doses, log-transformed IC50 values, and activity areas, which together form a single experimental record. The log-transformed IC50 values are especially important as they can be compared with predefined thresholds to determine the sensitivity or resistance of a cell line. After a series of preprocessing steps, we obtained data from the GDSC dataset for 561 cell lines and 222 drugs. The values in the cell line-drug response matrix are 0 and 1, representing resistance and sensitivity, respectively, with missing values filled as 0, resulting in a total of 100,572 drug responses. Using the same method and threshold setting, we obtained data from the CCLE dataset for 317 cancer cell lines and 24 drugs.

4.3. Comparison methods and evaluation metrics

To evaluate the performance of the ECACDR model in predicting drug response, we compare our method against two traditional ML models and the following comparison methods:

- **DeepCDR** [48] employs an innovative hybrid graph convolutional network technique that integrates multidimensional omics features of cancer cells and has the capability to automatically learn deep representations of drug chemical structures. This approach is used to predict cancer drug responses, aiming to provide a more reliable basis for precision treatment in cancer.
- **GraOmicDRP** [49] combines graph convolutional networks and convolutional neural networks to integrate drug and multi-omics data for predicting drug responses.
- **MOFGCN** [26] utilizes a unique graph convolutional network technique that integrates multi-omics data and considers linear correlation coefficients. This approach deeply explores the underlying features of complex heterogeneous networks formed by cell line and drug similarities.
- **GraphCDR** [25] employs an innovative graph neural network architecture combined with contrastive learning strategies. This model integrates various omics features of cancer cell lines, chemical structure information of drugs, and known cell line-drug response data, aiming to more accurately predict cancer drug responses.
- **NIHGCN** [27] employs an innovative heterogeneous GCN technique that considers interactions between neighboring elements. It integrates gene expression data from cell lines, molecular fingerprint features of drugs, and drug response information, while simultaneously accounting for interactions at both the node level and the component level during the modeling process.
- **GADRP** [50] constructs a drug-cell line pair-based DCP network, leveraging graph convolutional networks and autoencoders to predict cancer drug responses.
- **RedCDR** [50] proposes a parallel dual-branch architecture and constructs a dual-relational distillation model, which effectively integrates multi-omics data.

To comprehensively assess our model's performance, we use the following metrics:

$$AUC = \int_0^1 TPR(FPR)d(FPR), \tag{24}$$

where TPR (True Positive Rate) and FPR (False Positive Rate) are computed from the receiver operating characteristic (ROC) curve.

$$AUPR = \int_0^1 P(R)dR, \tag{25}$$

where P (Precision) and R (Recall) are derived from the precision–recall curve.

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}, \tag{26}$$

where TP, TN, FP, FN represent true positives, true negatives, false positives, and false negatives, respectively.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{27}$$

F1 Score balances precision and recall, ensuring that both false positives and false negatives are taken into account.

$$ACC = \frac{TP + TN}{TP + TN + FP + FN}. \tag{28}$$

Table 1
Performance comparison between comparison methods and the proposed method on GDSC and CCLE datasets. The best and second-best results are highlighted in bold and underline, respectively.

Dataset	Method	AUC (%)	AUPR (%)	F1 (%)	ACC (%)	MCC (%)
GDSC	Ridge regression	71.61 ± 0.55	28.26 ± 0.48	32.91 ± 0.73	75.16 ± 0.39	23.85 ± 0.63
	Random forest	75.04 ± 0.59	31.10 ± 0.81	35.22 ± 0.61	78.56 ± 0.48	26.16 ± 0.87
	DeepCDR	82.40 ± 0.43	48.53 ± 0.57	47.46 ± 0.80	87.45 ± 0.59	40.44 ± 0.94
	GraOmicDRP	78.16 ± 0.40	37.91 ± 0.79	41.19 ± 0.58	84.20 ± 0.66	32.93 ± 0.65
	MOFGCN	83.27 ± 0.23	49.45 ± 0.58	48.61 ± 0.73	87.64 ± 0.63	41.70 ± 0.88
	GraphCDR	82.96 ± 0.41	48.69 ± 1.33	48.07 ± 0.73	87.41 ± 0.32	41.00 ± 0.81
	NIHGCN	84.33 ± 0.36	51.66 ± 0.86	50.28 ± 0.71	88.01 ± 0.57	43.58 ± 0.81
	GADRP	82.20 ± 0.59	47.92 ± 1.32	47.09 ± 1.09	87.43 ± 0.66	40.03 ± 1.40
	RedCDR	<u>84.41 ± 0.46</u>	<u>52.48 ± 0.79</u>	<u>50.87 ± 0.80</u>	<u>88.24 ± 0.66</u>	<u>44.53 ± 1.04</u>
	Ours	**86.09 ± 0.32**	**53.62 ± 0.65**	**51.66 ± 0.77**	**89.33 ± 0.51**	**45.21 ± 0.81**
CCLE	Ridge regression	84.13 ± 0.47	75.66 ± 0.52	72.11 ± 0.94	81.33 ± 0.37	69.95 ± 0.65
	Random forest	86.09 ± 0.70	80.14 ± 1.13	76.24 ± 0.88	83.27 ± 0.75	71.16 ± 0.59
	DeepCDR	95.46 ± 0.56	86.39 ± 1.48	82.24 ± 1.80	93.24 ± 0.67	78.16 ± 2.19
	GraOmicDRP	95.49 ± 0.67	88.30 ± 1.41	82.08 ± 1.60	93.18 ± 0.71	77.99 ± 1.96
	MOFGCN	96.13 ± 0.65	88.93 ± 1.32	82.04 ± 1.82	92.91 ± 0.99	77.76 ± 2.33
	GraphCDR	94.83 ± 1.14	87.46 ± 1.80	82.53 ± 1.66	93.22 ± 0.74	78.40 ± 2.06
	NIHGCN	96.10 ± 0.70	89.25 ± 1.48	83.61 ± 1.65	93.57 ± 0.76	79.69 ± 2.05
	GADRP	96.28 ± 0.62	89.94 ± 1.17	83.47 ± 1.53	93.65 ± 0.77	79.64 ± 1.90
	RedCDR	<u>96.50 ± 0.32</u>	<u>90.26 ± 1.10</u>	<u>84.11 ± 1.94</u>	<u>93.88 ± 0.29</u>	<u>80.42 ± 2.72</u>
	Ours	**97.15 ± 0.07**	**91.47 ± 0.72**	**84.68 ± 0.49**	**94.08 ± 0.09**	**81.06 ± 0.56**

This metric measures the proportion of correctly predicted drug responses.

4.4. Experimental design

We conducted four different experimental tests to evaluate the performance of ECACDR.

Basic Evaluation Experiments for Drug Response Prediction (Test 1): To evaluate the drug response prediction ability of the proposed model, we first conducted basic evaluation experiments. Specifically, we split the drug response dataset into training, validation, and test sets with a ratio of 60%/20%/20%. The validation set was used for hyperparameter tuning, while the test set was used for final performance evaluation. To ensure the stability and robustness of the model, we conducted multiple experiments by varying the random seeds and assessed the model's performance on the validation and test sets using several metrics, including AUC, AUPR, F1 score, accuracy and MCC. The model with the highest AUC on the validation set was selected as the optimal model, and its performance on the test set was recorded.

New cell line/new drug prediction (Test 2): This test can evaluate the model's predictive ability for new cell lines or new drugs. In the cell line-drug association matrix, removing a row represents the loss of response data for a specific cell line, while removing a column represents the loss of response data for a specific drug. By using the removed data as the test set, it ensures that the cell lines or drugs appearing in the test set were not used during the training phase. This approach allows for an assessment of the model's predictive ability for new cell lines or new drugs.

Ablation study (Test 3): The test conducts an ablation study on our model to verify the contribution of different components to its performance. We performed the ablation experiments on five variants of the model:

- **w/o ef**: The edge feature construction is removed, and only node features are used for learning the cell lines.
- **w/o ca**: The complementary attention mechanism is removed.
- **w/o eu**: The external units used in the complementary attention module are removed.
- **w/o tu**: The topology updater module is removed.
- **w/o cl**: The complementary loss between node features and edge features is removed.

Case study (Test 4): Previous research indicates that there is missing data in some drug response datasets [25,27]. To validate the performance of our proposed model in predicting drug responses for novel cancer cell lines, we trained the model using all known drug response data from the GDSC dataset and predicted the unknown drug responses. Specifically, for each drug, we ranked its sensitivity probabilities and selected the 5 most sensitive drugs and the 5 least sensitive drugs. Since the true response values are unavailable, we further evaluated the accuracy of the model by performing a validation analysis on two clinically approved drugs, GSK690693 and Dasatinib, and referenced experimental evidence from relevant literature.

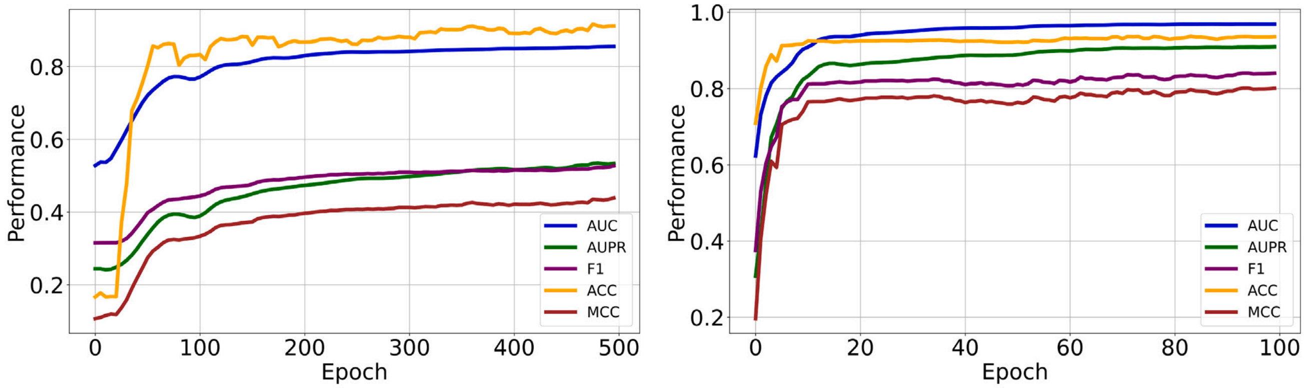
4.5. Experimental results

4.5.1. Basic evaluation experiments (Test 1)

Based on the experimental results shown in Table 1, our model outperforms other baseline models on multiple evaluation metrics, demonstrating superior performance. Our model shows significant improvements in metrics such as accuracy, precision, ACC, and F1 score compared to the second-best method. For example, the AUC is 1.68% higher than the second-best method, RedCDR, and the AUPR value is improved by 1.14%. This excellent performance can be attributed to the model's innovations in feature learning and fusion, particularly the introduction of edge features through the complementary attention mechanism, which allows the model to effectively learn the complex relationships between nodes and edges. This enables complementary and collaborative learning between features, better capturing the inherent patterns in biological data, thereby further improving prediction accuracy (see Fig. 2).

4.5.2. New cell line/new drug prediction (Test 2)

The test set was constructed by removing the response data of certain cell lines or drugs from the original data. This approach ensures that the test set contains cell lines or drugs that were not present during the training phase. Unlike traditional experiments, experiments with new cell lines and drugs better reflect the model's generalization ability when confronted with unseen data. Fig. 4 presents the results of our algorithm and other algorithms for predicting responses of new cell lines and drugs on the two datasets. From the figure, it is clear that, whether facing new cell lines or new drugs, our model ECACDR outperforms the comparison methods in terms of prediction accuracy on both the GDSC and CCLE datasets. From a biological perspective, the ability of our model to predict responses for new cell lines and drugs highlights its potential for uncovering novel therapeutic applications.



(a) GDSC (b) CCLE

Fig. 2. The figure shows the curve changes of five metrics: AUC, AUPR, F1, ACC and MCC over epochs, with the portion after stabilization not being displayed.

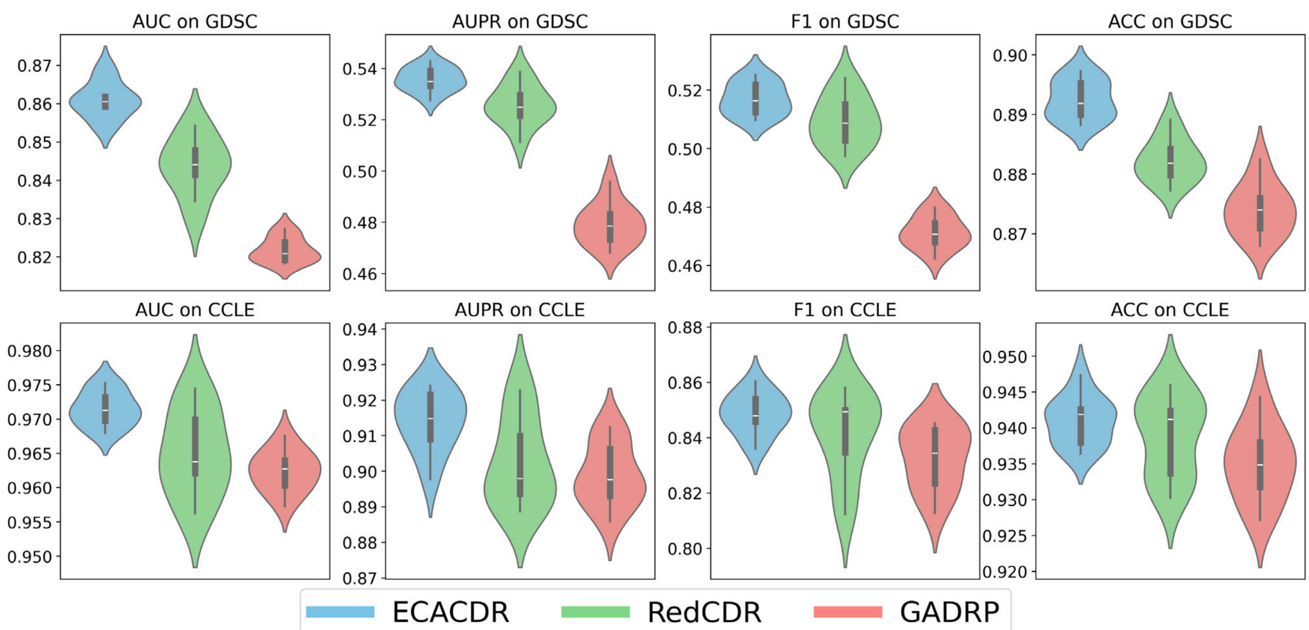


Fig. 3. Violin plots of our method and two comparison methods for four metrics.

Table 2
The results of the ablation experiments conducted to assess the contribution of each component in our model.

Dataset	Method	AUC (%)	AUPR (%)	F1 (%)	ACC (%)	MCC (%)
GDSC	w/o ef	85.29 ± 0.53	52.94 ± 0.41	50.66 ± 0.42	88.65 ± 0.71	44.36 ± 0.87
	w/o ca	85.39 ± 0.50	52.26 ± 0.44	51.24 ± 0.68	87.72 ± 0.59	43.88 ± 0.52
	w/o eu	85.13 ± 0.39	52.65 ± 0.45	51.17 ± 0.71	88.69 ± 0.74	44.71 ± 0.78
	w/o tu	85.67 ± 0.48	52.93 ± 0.98	51.08 ± 1.03	88.54 ± 0.47	44.62 ± 1.28
	w/o cl	85.66 ± 0.47	53.01 ± 0.67	50.94 ± 0.72	88.85 ± 0.60	44.70 ± 0.93
	ECACDR	**86.09** ± **0.32**	**53.62** ± **0.65**	**51.66** ± **0.77**	**89.33** ± **0.51**	**45.21** ± **0.81**
CCLE	w/o ef	96.42 ± 0.43	90.27 ± 1.09	83.83 ± 1.26	92.77 ± 0.89	79.61 ± 1.25
	w/o ca	96.57 ± 0.55	90.29 ± 1.20	83.94 ± 1.01	92.43 ± 0.66	79.60 ± 1.27
	w/o eu	96.32 ± 0.58	90.92 ± 0.95	83.71 ± 1.29	93.76 ± 0.83	80.85 ± 1.24
	w/o tu	96.81 ± 0.57	90.64 ± 1.23	83.99 ± 0.83	93.56 ± 0.24	80.70 ± 0.92
	w/o cl	96.89 ± 0.50	91.28 ± 0.82	84.18 ± 1.07	93.75 ± 0.29	80.82 ± 1.31
	ECACDR	**97.15** ± **0.07**	**91.47** ± 0.72	**84.68** ± **0.49**	**94.08** ± **0.09**	**81.06** ± **0.56**

The improved performance of ECACDR compared to baseline methods suggests that integrating edge features enhances the model's capability to generalize across different cancer types (see Fig. 3).

4.5.3. Ablation study (Test 3)

In this study, we conducted ablation experiments to verify the contribution of each component by systematically removing key parts and evaluating their impact on predictive performance. Based on the results in Table 2, the results confirm that every component is essential for improving overall performance.

w/o ef (without edge features): Removing edge feature construction and using only node features led to a significant decrease in performance when predicting responses for cell lines and drugs. This indicates

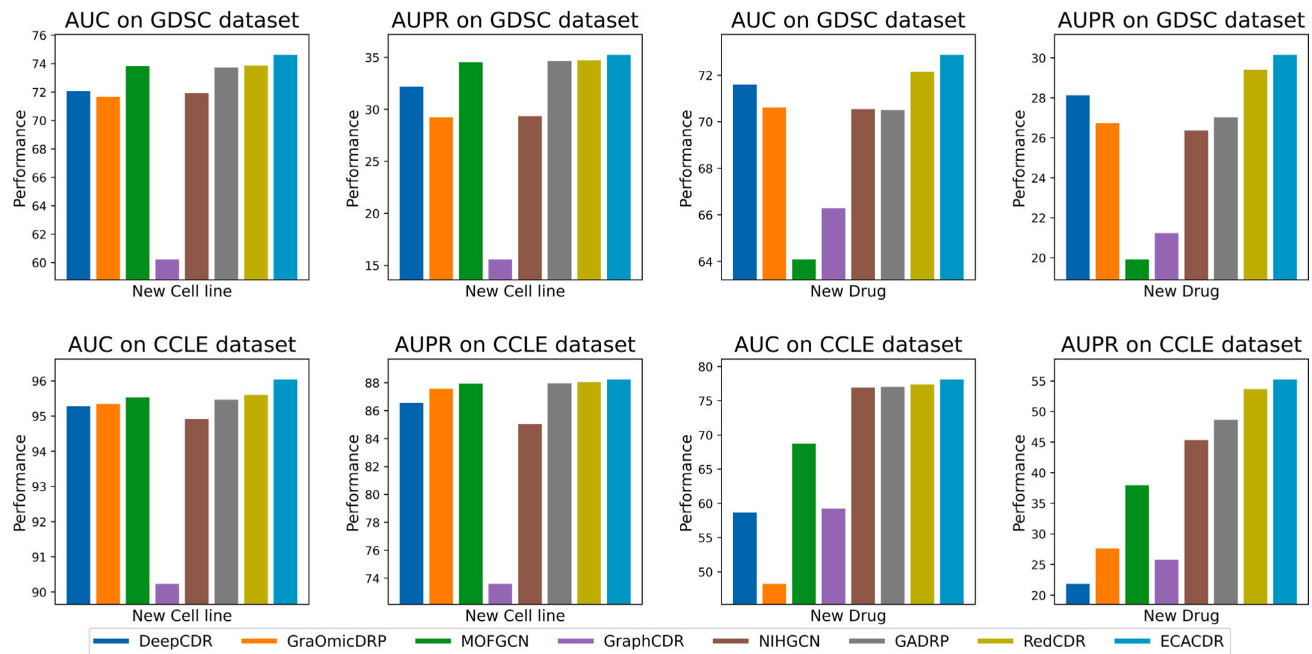


Fig. 4. The prediction performance on new drugs or new cell lines in the GDSC and CCLE datasets.

Table 3
The top 5 cell lines sensitive to the drugs Dasatinib and GSK690693 are listed, showing the cell line names and COSMIC IDs.

Drug	Rank	Cell line	COSMIC ID
Dasatinib	1	Hey	910784
	2	JURL-MK1	1479988
	3	8505C	924102
	4	A204	1327771
	5	786-0	1240149
GSK690693	1	RCH-ACV	1330994
	2	JEKO-1	1327765
	3	NCI-H929	724825
	4	MOLT-16	908147
	5	GA-10	1303896

that edge features are vital for capturing complex relationships and interactions between cell lines.

w/o ca (without complementary attention mechanism): Excluding the complementary attention mechanism resulted in diminished performance in learning the interactions between node and edge features. This highlights that the complementary attention mechanism is essential for effectively capturing and integrating the interaction between node and edge features, thereby improving the model's generalization to unseen data.

w/o eu (without external units): Removing the external units in the complementary attention module reduced the efficiency of information processing and contextual representation. The ablation results confirm that these external units significantly enhance the relevance and focus on different feature components, improving the model's capacity to interpret and integrate various feature types.

w/o tu (without topology updater module): Omitting the topology updater module constrained the model's ability to learn and adapt to the structural features of the data. The effectiveness of this module lies in its capacity to dynamically adjust the graph structure, capturing and reflecting the relationships between samples more accurately and improving prediction accuracy.

w/o cl (without complementary loss): Removing the complementary loss function between node and edge features led to a decline in feature integration performance. The complementary loss function acts as an additional constraint that promotes joint learning of node and edge features, enabling the model to better utilize the complementary information between these features.

The results from these experiments clearly indicate that each component in the model plays a unique and critical role in enhancing performance. The removal of any component results in a significant decrease in performance, thus validating the necessity and effectiveness of all components in the model.

4.5.4. Case study (Test 4)

Table 3 lists the top 5 cell lines predicted by our model to be most sensitive to two classic clinical drugs, along with their COSMIC IDs.

HEY, JURL-MK1, and 786-0 have been identified as the cell lines most sensitive to Dasatinib. This prediction is supported by previous research: Le et al. [51] observed that Dasatinib induces limited apoptosis and significant autophagy inhibition in HEY ovarian cancer cells, thereby suppressing cell growth. The sensitivity of cell lines BCR-ABL1 and 786-0 to Dasatinib is also validated by studies from Obr [52], Roseweir [53], and others. RCH-ACV, JEKO1, and MOLT-16 have been identified as the cell lines most sensitive to GSK690693. The observed sensitivity is likely due to the hyperactivation of the PI3K/AKT pathway in these cell lines, which plays a crucial role in tumor survival and resistance to apoptosis. GSK690693 acts by targeting AKT phosphorylation, thereby disrupting downstream pro-survival signaling and leading to growth inhibition and apoptosis [54,55].

The study of these specific drug cases demonstrates that our model possesses a significant capability for predicting clinical drug responses. The model's predictions, consistent with previous findings, underscore its potential for identifying new drug sensitivities, offering valuable insights for personalized medicine and drug development.

5. Conclusion

In summary, the proposed model ECACDR successfully addresses the limitations of traditional drug response prediction methods in capturing inter-cell line relationships and achieving feature complementarity by introducing edge features and a node-edge complementary attention mechanism. Additionally, the topology update module dynamically aggregates neighborhood information, further optimizing the learning process of node features and achieving deep integration of multi-omics data while maintaining feature independence and complementarity. These innovations lead to remarkable improvements in prediction performance which are confirmed through exceptional outcomes achieved using the GDSC and CCLE datasets. In addition to its predictive accuracy, the ECACDR model enhances biological interpretability by incorporating inter-cell line relationships and multi-omics data. Furthermore, the model demonstrates potential for clinical applications, providing strong support for the development of personalized treatment plans and optimized drug selection.

CRediT authorship contribution statement

Chuang Li: Writing – original draft, Methodology. **Minhui Wang:** Investigation, Data curation. **Chang Tang:** Writing – review & editing, Supervision. **Yanfeng Zhu:** Resources, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The source code is available at https://github.com/khalilclee/ECACDR.

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